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Prototype-Aware Reliable Target Structure Learning for Cross-Domain Few-Shot Remote Sensing Image Classification

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***Abstract*—Cross-domain few-shot image classification aims to learn transferable representations from a labeled source domain and enable few-shot recognition in a target domain. Recent studies improve cross-domain representation learning by incorporating unlabeled target data into semi-supervised training with labeled source data. However, most of them mainly exploit unlabeled target samples through prediction-level consistency, while lacking an explicit mechanism to organize target-domain samples into stable structural relations. This may lead to ambiguous feature distributions and unstable decision boundaries under large domain shifts. To address this issue, we propose Prototype-Aware Reliable Target Structure Learning (PARTS), a framework for cross-domain few-shot remote-sensing image classification. PARTS consists of two complementary components: Prototype-Guided Target Structure Learning (PTSL) and Prototype Reliability Structure Consolidation (PRSC). PTSL learns latent target-domain structures from unlabeled samples by organizing them through prototype-induced relations and refining their representations via prototype-guided residual adaptation. PRSC further consolidates the learned target structure by estimating prototype reliability and reducing redundancy among active prototypes. Together, the two components promote stable, compact, and discriminative target-domain representations under large domain shifts. Extensive experiments on multiple cross-domain remote-sensing benchmarks demonstrate that PARTS achieves competitive performance under both 1-shot and 5-shot settings.**

***Index Terms*—Cross-domain few-shot learning, prototype learning, semi-supervised learning, target-domain adaptation.**

I. Introduction

REMOTE sensing (RS) image classification has achieved remarkable progress with deep neural networks and large-scale annotated datasets [1], [2]. However, in practical RS applications, collecting sufficient labeled data for every new target domain is expensive and often requires expert knowledge [3]. Meanwhile, RS images acquired from different sensors, platforms, spatial resolutions, seasons, and geographic regions usually suffer from significant domain shifts [4], [5], [6]. These factors make cross-domain few-shot learning (CDFSL) particularly important for RS applications, where models are expected to transfer knowledge from a labeled source domain to novel target-domain tasks with only a few labeled samples [3], [7].

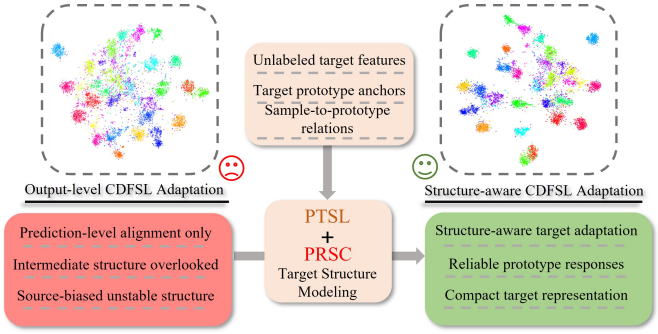


Fig. 1. Motivation of the proposed structure-aware CDFSL adaptation. Unlike output-level alignment that may ignore intermediate target structures, our method enhances target structure modeling and yields more compact and discriminative representations.

According to existing cross-domain few-shot learning studies, current methods can be roughly divided into meta-learning-based and transfer-learning-based approaches [7]. Meta-learning methods learn task-level transferable knowledge from source-domain episodes, aiming to rapidly adapt to novel few-shot tasks [8], [9], [10], [11], [12], [13]. Representative methods improve cross-domain generalization by exploiting local–global semantic relationships, adversarial style perturbation, domain mapping, or prompt-based adaptation [12], [13], [14], [15], [16]. In RS, recent CDFSL studies further investigate few-shot transformation learning, global-to-local channel attention, and geographical prior guidance to improve cross-domain adaptation under complex imaging conditions [17], [18], [19]. These methods are effective in capturing task-level regularities and improving adaptation to unseen categories. However, RS scenes usually exhibit more complex visual variations than natural images, including sensor-dependent imaging characteristics, heterogeneous land-cover distributions, and large changes in spatial layout and acquisition conditions [4], [5], [6]. Since meta-training episodes are usually sampled from source or base domains, the learned task-level knowledge

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can still be biased toward source-domain structures. As a result, these methods may suffer when transferred to target RS domains with substantially different spatial layouts, scales, textures, and imaging conditions.

Transfer-learning-based methods follow the feature reuse paradigm, where generalized representations are first learned from source-domain data and then adapted to target-domain few-shot tasks [20], [21]. Recent studies further exploit unlabeled target-domain data through semi-supervised learning, pseudo-labeling, distillation, or consistency-based learning, demonstrating that target samples can provide useful adaptation cues for cross-domain few-shot learning [22], [23], [24], [25]. By incorporating unlabeled target data during representation learning, these methods alleviate source-only overfitting and improve adaptation to novel domains. Nevertheless, their target-domain supervision is usually derived from individual pseudo-labels or weak–strong view consistency, which mainly regularizes each target sample independently. As a result, the broader structural organization of the unlabeled target domain remains insufficiently modeled.

This limitation becomes more pronounced in remote sensing CDFSL, where target scenes often exhibit complex spatial layouts, large intra-class variations, and severe distribution shifts across datasets [4], [5], [6]. In such cases, sample-level consistency can stabilize predictions but may fail to recover reliable relationships among target samples. As shown in Fig. 1, the baseline can form several local target clusters, but the learned target-domain representation is still fragmented and less compact. This motivates us to move beyond sample-level consistency and explicitly model reliable target-domain structures.

Prototype-based few-shot learning provides a natural way to represent categories or latent structures with prototype anchors [9], [10], [12]. This is particularly appealing for remote sensing CDFSL, since prototypes can serve as compact structural references for organizing target samples under limited supervision. However, directly constructing reliable prototypes in an unlabeled target domain is challenging. Due to severe domain shifts and large intra-class variations in RS imagery [4], [5], [6], target prototypes estimated from noisy assignments may be unstable, under-used, or redundant. If such prototypes are treated equally during adaptation, unreliable prototype responses may propagate incorrect structural information and further disturb target feature refinement. Therefore, the key problem is not only how to build target-domain prototypes, but also how to identify reliable prototypes that can serve as stable structural anchors for target-domain representation learning.

To address the above issues, we propose Prototype-Aware Reliable Target Structure Learning (PARTS) for cross-domain few-shot RS image classification. PARTS consists of two core components: Prototype-Guided Target Structure Learning (PTSL) and Prototype Reliability Structure Consolidation (PRSC). PTSL learns target structure through a set of adaptive prototypes and the relations between target samples and these prototypes. We refer to the resulting prototype-based structural representation as the target prototype field. Based on these relations, PTSL aggregates prototype-level structural information and injects it back into the target feature through residual refinement. In this way, target adaptation is no longer limited to enforcing consistency between two augmented views of the same image, but is further guided by prototype-induced relationships among different target samples.

Since target prototypes learned from unlabeled data may not be equally reliable, PRSC further consolidates the learned target structure by estimating prototype reliability from target usage and weak–strong response consistency. The estimated reliability is used to reweight prototype responses, so that stable and frequently activated prototypes are emphasized while unreliable prototype-induced structures are suppressed. In addition, active prototype de-redundancy is introduced to reduce redundancy among frequently activated prototypes and encourage the prototype field to cover more diverse target structures. As illustrated in Fig. 1, our method organizes target samples around reliable target prototypes, leading to more compact and better separated target-domain representations.

We evaluate the proposed method on multiple cross-domain RS few-shot benchmarks. The results show consistent improvements under both 5-way 1-shot and 5-way 5-shot settings, while ablation and visualization analyses further support the roles of PTSL and PRSC. Based on the above design, the main contributions of this work are summarized as follows:

1) We propose PARTS to model relationships among unlabeled target samples overlooked by sample-level consistency. PARTS organizes target representations around reliable prototypes, producing compact and discriminative structures for cross-domain few-shot transfer.
2) We design PTSL to organize unlabeled target samples with prototypes and use the resulting structure to refine target representations, thereby providing structural information that is missing from the original target features.
3) We introduce PRSC, which estimates prototype reliability from prototype usage and weak–strong response consistency, and regularizes frequently activated prototypes to reduce redundant target assignments.

The remainder of this article is organized as follows. Section II reviews related work on cross-domain few-shot learning, semi-supervised cross-domain representation learning, and prototype-based structure learning. Section III presents the proposed PARTS framework, including PTSL and PRSC, in detail. Section IV describes the datasets, experimental setup, and result analysis. Finally, Section V concludes this article.

II. Related Work

A. Cross-Domain Few-Shot Learning

CDFSL aims to transfer knowledge from a source domain to novel classes in a target domain, where only limited labeled samples are available and the source–target domain gap is usually significant. Since Guo et al. introduced the broader CDFSL benchmark [7], this problem has attracted increasing attention in few-shot learning. Early CDFSL studies mainly followed the meta-learning paradigm, where episodic tasks are constructed during training to improve rapid adaptation to unseen domains [11], [26], [27]. Later studies showed that transfer-learning-based pipelines with strong pretraining and target-domain fine-tuning can outperform early meta-learning

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methods in several cross-domain settings [20], [21], [28]. Beyond these training paradigms, subsequent methods further improve cross-domain generalization through data transformation, feature adaptation, semantic alignment, or relation modeling. For example, FWT introduces affine transformation layers to enhance feature transferability [11], NSAE uses noise-enhanced autoencoders to enlarge feature distribution variations [29], TSA adapts features with additional parameter adapters [27], and TA2-NET estimates task-specific parameters for different cross-domain tasks [30].

Remote sensing CDFSL has recently attracted increasing attention because large domain gaps and scarce target labels commonly exist across RS datasets. Existing RS-related few-shot and cross-domain methods have explored different imaging modalities and adaptation strategies [31]. For hyperspectral image classification, CF-Trans learns intra-domain distinctiveness and inter-domain invariance with attentive and adversarial networks [17], while Gprior-FSN generates geospatially and spectrally correlated pseudo-samples using geographical priors [19]. For RS-oriented CDFSL and scene-level recognition, UDL improves knowledge transfer by sample partitioning [25], DMN aligns source and target distributions with dual autoencoders [15], MVP introduces prompt tuning for RS adaptation [16], and CDML designs a dual-stage cross-domain meta-learning task with learnable affine transformation for few-shot RS object classification [32]. More recently, DMPC employs dual-domain representation modeling and pseudo-prototype learning [33]. Although these methods improve few-shot transfer in RS scenarios, they mainly focus on domain alignment, support-set prototype refinement, or discriminative representation learning, while relationships among unlabeled target samples remain insufficiently explored.

B. Semi-Supervised Cross-Domain Representation Learning

Semi-supervised learning (SSL) exploits unlabeled data mainly through pseudo-labeling and consistency regularization [34], [35], [36]. Representative methods include Noisy Student for iterative teacher–student pseudo-label refinement [37], Mean Teacher for stable exponential moving average (EMA) teacher supervision [35], UDA for weak-to-strong augmentation consistency [38], and FixMatch for confidence-based pseudo-labeling with consistency regularization [39]. These techniques have been further introduced into cross-domain representation learning, where unlabeled target samples are exploited to alleviate source–target distribution shifts [23], [40]. In semi-supervised domain adaptation, MCL regularizes target samples from inter-domain, intra-domain, and sample-level views through prototype-based optimal transport, class-wise contrastive clustering, and consistency-based self-training [41], while ProML introduces prototype-based multi-level learning with intra-domain pseudo-label aggregation, cross-domain prototype alignment, and dual consistency regularization [42]. In the CDFSL setting, DDN performs dynamic distillation by matching weakly augmented teacher predictions with strongly augmented student predictions on unlabeled data [23]. In RS, BSCA performs semi-supervised bidirectional sample-class alignment for cross-domain scene classification, where supervised and unsupervised alignment strategies are jointly used to reduce domain shift [40]. Although these methods improve the utilization of unlabeled target data, they are still mainly developed around pseudo-label reliability, prediction consistency, or class-level alignment, leaving fine-grained target-domain structural relations insufficiently modeled.

C. Prototype-Based Structure Learning

Prototype-based learning has been widely adopted in few-shot classification because prototypes provide compact and interpretable representations for novel classes [43]. ProtoNet represents each class by the mean feature of its support samples and classifies query samples according to their distances to class prototypes [10]. Related metric-based methods further show that sample relations and learned similarity metrics are important for few-shot recognition [9], [44]. In CDFSL, prototype and relation information has been exploited to improve cross-domain transfer [45]. For example, Dara maintains invariant prototype relations under domain shifts to alleviate the covariate distribution gap [46], LDP-Net integrates local and global semantic relations to enhance transferable representations [12], and DMPC employs pseudo-prototype learning to improve cross-domain representation robustness and discriminability [33]. Prototype information has also been used in semi-supervised domain adaptation, such as prototype-based optimal transport in MCL [41] and cross-domain prototype alignment in ProML [42]. Although these methods demonstrate the effectiveness of prototype-based representation learning, most of them construct prototypes from labeled support samples, pseudo-labeled target samples, or source-domain categories. Under unlabeled target-domain adaptation, such prototypes may be affected by noisy assignments and unstable target distributions, making it difficult to capture reliable target-domain structures.

MCL uses prototype-based optimal transport for domain alignment [41], while ProML employs target prototypes for pseudo-label aggregation and cross-domain alignment [42]. In contrast to class-associated prototype formulations, PARTS constructs a category-agnostic field whose prototypes serve as structural anchors rather than class representatives. Their responses guide residual refinement of target features. Furthermore, prototype usage and weak–strong response consistency jointly determine reliability, which modulates both response weighting and prototype memory updates. The distinctive design lies in coupling category-agnostic structural refinement with prototype-level reliability regulation during target adaptation.

III. Methodology

In this section, the proposed framework is described in detail. The remainder of this section is organized into five parts: 1) Preliminaries; 2) Overview; 3) Prototype-Guided Target Structure Learning; 4) Prototype Reliability Structure Consolidation; and 5) Overall optimization and target-domain evaluation.

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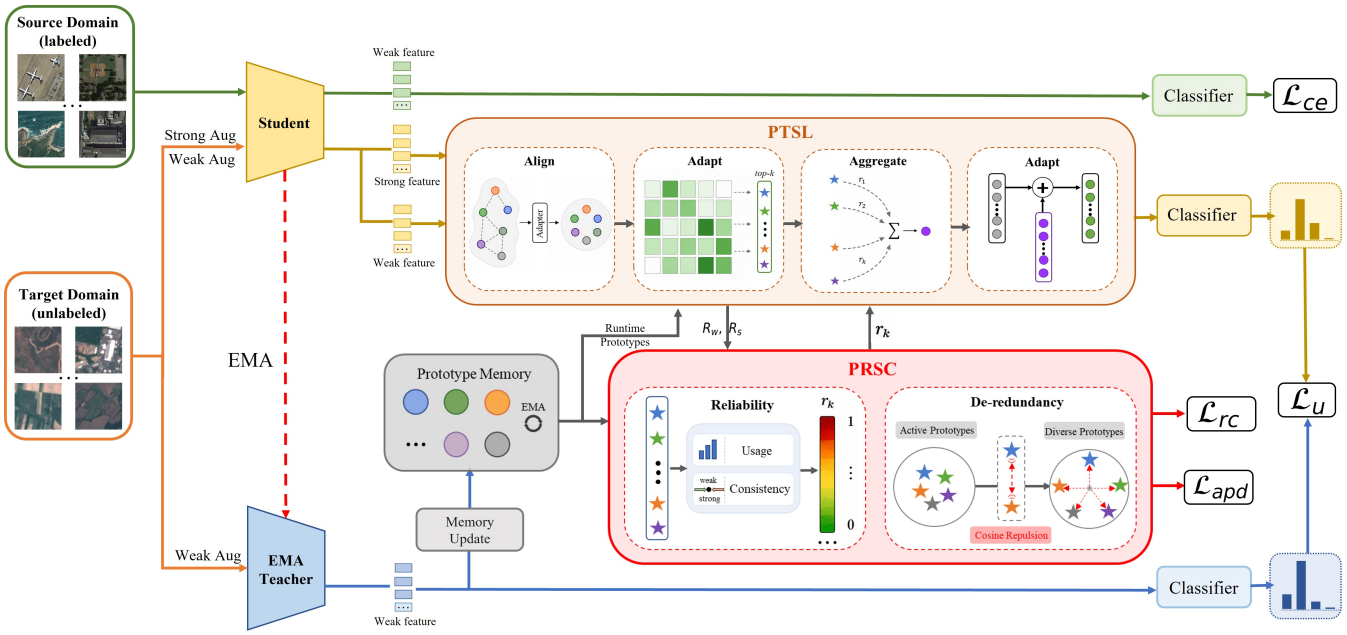


Fig. 2. Overview of the proposed PARTS framework. PTSL learns target structure using adaptive prototypes for unlabeled target samples, while PRSC consolidates the learned structure through reliability estimation and active prototype de-redundancy.

A. Preliminaries

We study cross-domain few-shot remote-sensing classification with unlabeled target-domain data. Given a labeled source dataset $D_S = \{(x_i^s, y_i^s)\}_{i=1}^{N_S}$ and a target domain T, we follow STARTUP [22] and DDN [23] by splitting target images into an unlabeled subset D_T^U and an evaluation subset D_T^E, with $D_T = D_T^U \cup D_T^E$. The unlabeled subset $D_T^U = \{x_i^t\}_{i=1}^{N_T^U}$ contains 20% of target images and is used without labels for representation learning, while the remaining 80% forms D_T^E for episodic few-shot evaluation.

The learning procedure consists of two stages. In the cross-domain representation learning stage, the model is trained with D_S and the unlabeled images in D_T^U. The source data provide supervised classification signals, while D_T^U provides unlabeled target-domain distribution information for teacher–student consistency learning and target prototype field construction. In the target-domain few-shot adaptation and evaluation stage, standard N-way K-shot episodes are sampled only from D_T^E. Each episode contains a support set S_T with K labeled samples from each of the N selected classes and a query set Q_T from the same classes. The model is adapted using S_T and evaluated on Q_T.

B. Overview

In the cross-domain representation learning stage, both the labeled source data from D_S and the unlabeled target data from D_T^U are involved in training. As shown in Fig. 2, the model follows a teacher–student training scheme. The source branch is supervised by source labels to maintain category discrimination, whereas the target branch learns from unlabeled target images through weak–strong consistency. To make the target learning process less dependent on prediction confidence alone, we introduce a target prototype field to describe how target samples are distributed around a set of adaptive prototypes.

The prototype field is category-agnostic: prototypes are not assigned class identities, and their number is independent of the number of target classes. Unlike class centers constructed by grouping samples according to predicted labels, these structural anchors receive contributions through feature-based soft assignments. An incorrect class pseudo-label therefore does not directly determine which prototype receives a sample's contribution. This avoids directly incorporating class-label errors into prototype aggregation, while allowing each sample to contribute to multiple anchors. The field can nevertheless be affected by inaccurate features and assignments, including the indirect influence of pseudo-label supervision on the backbone. PRSC addresses this remaining uncertainty by using prototype usage and weak–strong consistency to regulate prototype contributions and memory updates.

The target branch is built around PTSL and PRSC. PTSL, illustrated in Fig. 3, maps target features to a prototype-aligned space, assigns them to runtime prototypes with sparse responses, and uses the aggregated prototype information to refine target features through residual adaptation. PRSC, shown in Fig. 4, further calibrates this prototype field by estimating prototype reliability from usage and weak–strong response consistency. The reliability scores are used to guide response consolidation and structural consistency, while frequently activated runtime prototypes are regularized to reduce redundancy. The training objective combines source supervision, target consistency, reliability-guided structure consistency, and active prototype de-redundancy. After representation learning, few-shot episodes are sampled from the held-out target evaluation subset for adaptation and testing under the standard N-way K-shot protocol.

C. Prototype-Guided Target Structure Learning

PTSL learns target structure by organizing unlabeled target samples around a set of adaptive prototypes. Instead of

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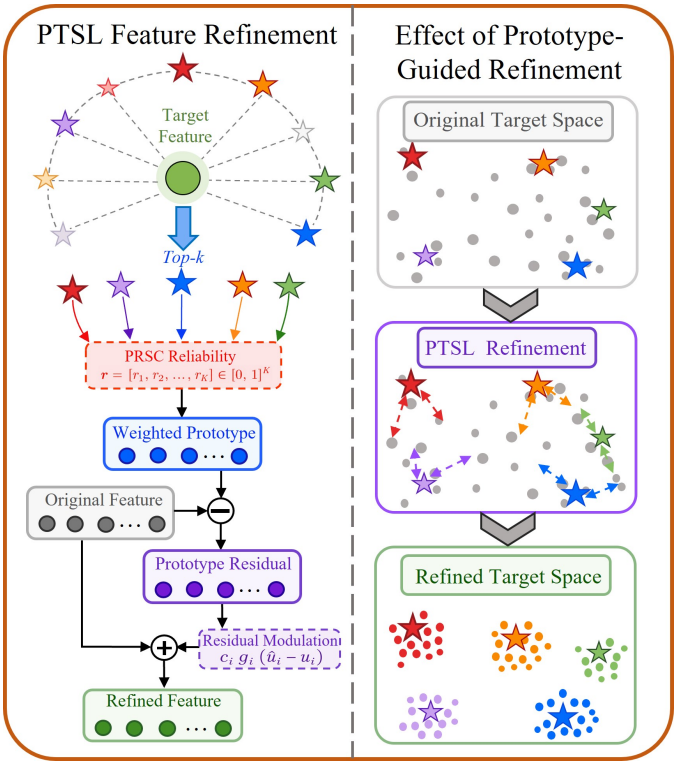


Fig. 3. Overview of PTSL. Target features are assigned to top-k runtime prototypes, aggregated into prototype context, and refined by confidence-controlled residual adaptation.

applying teacher–student consistency directly on the original target features, PTSL first describes each target sample by its relation to a set of adaptive prototypes and then uses the prototype-induced structure to refine the target representation. In this way, target-domain adaptation is guided not only by category-level pseudo-labels, but also by the local structure formed between target samples and prototypes.

Given a target image x_i^t, the student backbone $f_\theta(\cdot)$ extracts its raw feature as

$$z_i = f_\theta(x_i^t). \tag{1}$$

PTSL introduces a raw-to-prototype adapter $A(\cdot)$ to map the backbone feature into a compact prototype space:

$$u_i = \mathrm{Norm}(A(z_i)), \tag{2}$$

where $\mathrm{Norm}(\cdot)$ denotes ℓ_2 normalization. The adapter is a bias-free linear mapping $A(z_i) = W_A z_i$, with $W_A \in \mathbb{R}^{256\times512}$ initialized with random orthonormal rows. The mapping weights remain fixed during training, while gradients propagate through the adapter to update the student backbone. The resulting feature u_i provides a unified 256-dimensional representation for prototype assignment, prototype memory update, and residual adaptation, keeping these operations in the same coordinate space. The fixed projection provides a compact and stable coordinate system for prototype assignment and online memory accumulation without introducing additional trainable projection parameters. Its fixed basis also maintains a consistent feature space for accumulating target-domain prototype statistics across training iterations.

PTSL maintains M learnable prototypes $P = \{p_k\}_{k=1}^M$ and an online prototype memory $\mathcal{M} = \{m_k\}_{k=1}^M$. The learnable prototypes serve as trainable structural anchors, while the memory stores target-domain prototype statistics accumulated during training. At each iteration, the runtime prototype is obtained by blending the learnable prototype with its memory counterpart:

$$p_k^{rt} = \mathrm{Norm}\left((1-\beta)p_k + \beta m_k\right), \tag{3}$$

where β is the memory blending coefficient. The runtime prototypes $\{p_k^{rt}\}_{k=1}^M$ form the current adaptive prototype set.

For each target feature u_i, PTSL first computes its response to the runtime prototypes by cosine similarity followed by temperature-scaled softmax:

$$a_{ik} = \frac{\exp(\cos(u_i, p_k^{rt})/\tau)}{\sum_{j=1}^M \exp(\cos(u_i, p_j^{rt})/\tau)}, \tag{4}$$

where τ is the assignment temperature. To encourage local prototype competition and avoid overly diffuse responses, only the top-k responses are retained and renormalized:

$$\tilde{a}_{ik} = \frac{a_{ik}\mathbf{1}[k \in \mathrm{Top}_k(\mathbf{a}_i)]}{\sum_{j\in\mathrm{Top}_k(\mathbf{a}_i)} a_{ij} + \epsilon}, \tag{5}$$

where ϵ is a small constant for numerical stability. In practice, only the top-ranked prototype responses are retained to obtain a sparse assignment. Together, the runtime prototype set $\{p_k^{rt}\}_{k=1}^M$ and the sparse sample-to-prototype response vector $\tilde{\mathbf{a}}_i$ define the target prototype field. The prototypes serve as adaptive structural anchors, while $\tilde{\mathbf{a}}_i$ describes the relation of each target sample to these anchors.

Since the prototype field is constructed from unlabeled target samples, not all prototype responses are equally reliable. PTSL therefore calibrates the sparse prototype response using the prototype reliability estimated by PRSC. Let $\tilde{a}_{ik}^r$ denote the reliability-aware response of sample i to prototype k. This calibrated response is used as the effective response for prototype aggregation and residual adaptation. Specifically, PTSL aggregates the runtime prototypes as

$$\hat{u}_i = \sum_{k=1}^M \tilde{a}_{ik}^r p_k^{rt}. \tag{6}$$

The aggregated representation $\hat{u}_i$ can be viewed as the prototype-guided context feature of u_i in the target prototype field. Therefore, PTSL does not refine target features according to raw assignment scores alone. Instead, reliable prototypes contribute more to the target structural representation, while unreliable prototypes are suppressed.

To prevent unreliable target samples from receiving overly strong prototype adaptation, we estimate a sample-level teacher confidence gate from the EMA teacher prediction on the weak target view:

$$c_i = 1 - \frac{H\left(\mathrm{Softmax}\left(h_{\theta'}(f_{\theta'}(x_i^{t,w}))\right)\right)}{\log C}, \tag{7}$$

where $f_{\theta'}(\cdot)$ and $h_{\theta'}(\cdot)$ denote the EMA teacher backbone and classifier, $H(\cdot)$ is the entropy function, and C is the number of classes. A lower prediction entropy indicates a more reliable target sample and therefore yields a larger confidence gate.

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Although $\hat{u}_i$ captures prototype-based structural information, it is an aggregate of prototypes constructed from unlabeled target data and does not explicitly preserve sample-specific discriminative information. Directly replacing u_i with $\hat{u}_i$ would discard the original feature and increase sensitivity to assignment noise.

To regulate the contribution of prototype guidance, PTSL computes a sample-specific residual gate from the student backbone feature and the reliability-aware prototype response. The response is first embedded as

$$\mathbf{e}_i = W_c \tilde{\mathbf{a}}_i^r + \mathbf{b}_c \in \mathbb{R}^d. \tag{8}$$

The embedding is concatenated with the backbone feature and passed through a two-layer gate network:

$$\mathbf{G}_i = s_t \, \sigma(W_2 \, \mathrm{GELU}(W_1[z_i; \mathbf{e}_i] + \mathbf{b}_1) + \mathbf{b}_2), \tag{9}$$

where $[\cdot;\cdot]$ denotes concatenation, σ is the sigmoid function, and s_t is the target-field strength at training iteration t. The network has dimensions $2d \to d \to d$, with $d = 512$. The channel-wise outputs are averaged to obtain the scalar gate:

$$g_i = \frac{1}{d} \sum_{c=1}^{d} G_{i,c}. \tag{10}$$

The gate parameters are optimized through the target teacher–student consistency loss $\mathcal{L}_u$, without an additional gate-specific loss.

Using this gate, PTSL incorporates prototype guidance through a residual update:

$$\tilde{u}_i = u_i + \omega(e) c_i g_i \, (\hat{u}_i - u_i), \tag{11}$$

where $\omega(e)$ is the PTSL warm-up weight at epoch e. The residual form moves u_i toward the prototype context rather than replacing it. The factors c_i, g_i, and $\omega(e)$ control the movement, allowing reliable samples to receive stronger prototype guidance while keeping uncertain samples close to their original representations. The refined representation is mapped back to the backbone feature space as $\tilde{z}_i = R_\psi(\tilde{u}_i)$, where $R_\psi : \mathbb{R}^{256} \to \mathbb{R}^{512}$ is a learnable linear reconstruction layer. The resulting $\tilde{z}_i$ is fed into the shared classifier for target-domain teacher–student consistency learning.

The prototype memory is updated online using weak target features extracted by the EMA teacher. Given the weak target view $x_i^{t,w}$, we first map the teacher feature into the prototype-aligned space:

$$z_{i,T}^w = \mathrm{Norm}\left(A\left(f_{\theta'}(x_i^{t,w})\right)\right), \tag{12}$$

where $f_{\theta'}(\cdot)$ is the EMA teacher backbone and $A(\cdot)$ denotes the raw-to-prototype adapter. Let ρ_{ik}^w be the effective assignment weight of sample i to prototype k, which combines the sparse weak-view prototype response and the response confidence. The batch-level memory estimate for prototype k is then computed by weighted aggregation:

$$\bar{m}_k = \mathrm{Norm}\left(\frac{\sum_i \rho_{ik}^w z_{i,T}^w}{\sum_i \rho_{ik}^w + \epsilon}\right). \tag{13}$$

The prototype memory is updated with an EMA rule:

$$m_k \leftarrow \mathrm{Norm}\left((1-\eta_k) m_k + \eta_k \bar{m}_k\right), \tag{14}$$

where η_k is the effective update rate for prototype k. In our framework, η_k is modulated by the prototype reliability estimated by PRSC, so reliable prototypes are updated more actively, while unreliable prototypes are updated conservatively.

In the target branch, the weak view provides teacher predictions, confidence estimation, prototype memory updates, and weak-side prototype responses. The strong view is refined by PTSL using the PRSC-calibrated prototype response and is used for the final target consistency prediction. This design allows PTSL to exploit stable teacher-side structural information while applying reliability-aware prototype-guided feature adaptation to the student branch.

D. Prototype Reliability Structure Consolidation

PRSC is introduced to consolidate the prototype-guided target structure learned by PTSL. Rather than acting as an isolated regularizer, PRSC is coupled with PTSL through the prototype responses. PTSL first produces sparse responses between target samples and runtime prototypes, while PRSC estimates prototype reliability from weak–strong response statistics and feeds reliability-aware responses back to PTSL for prototype aggregation and residual adaptation. For prototype memory update, the teacher weak-view assignment is used, while the reliability score modulates the effective update rate. In addition, PRSC regularizes frequently activated runtime prototypes to reduce redundancy in the target prototype field.

For each unlabeled target image x_i^t, we generate a weak view $x_i^{t,w}$ and a strong view $x_i^{t,s}$. The student backbone first extracts their target features, which are then mapped to the prototype-aligned space and assigned to the runtime prototypes in PTSL. This produces the sparse prototype responses $\tilde{\mathbf{a}}_i^w$ and $\tilde{\mathbf{a}}_i^s$, where $\tilde{a}_{ik}^w$ and $\tilde{a}_{ik}^s$ denote the responses of prototype k for the weak and strong views, respectively. PRSC estimates prototype reliability from these responses rather than raw image features, because the responses directly describe how target samples are organized in the current prototype field.

In the final configuration, PRSC uses two prototype-level statistics: usage and weak–strong consistency. The usage of prototype k is defined as the average response mass assigned to it:

$$U_k = \frac{1}{B} \sum_{i=1}^{B} \frac{\tilde{a}_{ik}^w + \tilde{a}_{ik}^s}{2}, \tag{15}$$

where B is the target batch size. A larger usage value indicates that the prototype is more frequently involved in representing target samples.

The weak–strong consistency of prototype k measures whether the two augmented views activate this prototype in a similar manner:

$$C_k = 1 - \frac{\sum_{i=1}^{B} |\tilde{a}_{ik}^w - \tilde{a}_{ik}^s|}{\sum_{i=1}^{B} (\tilde{a}_{ik}^w + \tilde{a}_{ik}^s) + \epsilon}, \tag{16}$$

where ϵ is a small constant for numerical stability. A prototype receives a high consistency score when it responds similarly

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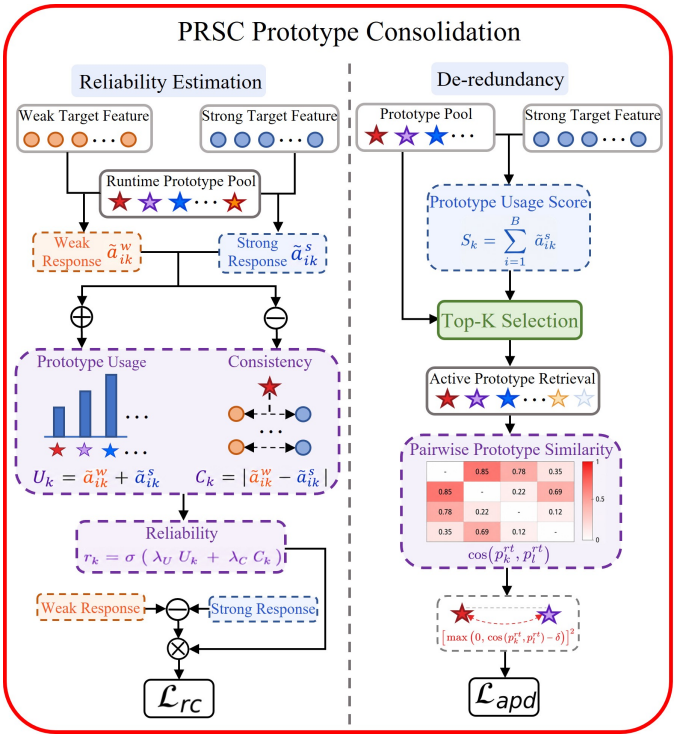


Fig. 4. Overview of PRSC. Prototype reliability is estimated from usage and weak–strong response consistency, and is used to guide response consolidation and structural consistency. Active prototype de-redundancy further regularizes frequently activated runtime prototypes.

to weak and strong views, while prototypes with unstable responses obtain lower consistency scores.

The usage and consistency statistics are first smoothed using an EMA with a momentum of 0.97. For each statistic, the mean and standard deviation across prototypes are tracked using a second EMA with a momentum of 0.95. The smoothed statistics are standardized by subtracting the corresponding running mean and dividing by the running standard deviation, with a minimum denominator of 10^{-6}, and then clipped to $[-6, 6]$. Both EMA stages are initialized from the first batch. Based on the normalized statistics, the reliability score of prototype k is computed as

$$r_k = \sigma\left(\lambda_U \bar{U}_k + \lambda_C \bar{C}_k\right), \tag{17}$$

where $\bar{U}_k$ and $\bar{C}_k$ denote the normalized usage and consistency statistics, λ_U and λ_C are their weights, and $\sigma(\cdot)$ is the sigmoid function. The reliability score jointly reflects prototype usage and weak–strong assignment consistency. Sample confidence c_i measures certainty in the teacher's class prediction, whereas prototype reliability r_k summarizes the usage and cross-view stability of structural assignments. Reliability is estimated from prototype responses without confidence-based weighting or filtering, so class uncertainty and structural consistency need not coincide. Individual assignments contribute to the batch statistics and their temporal averages, linking sample behavior to prototype reliability. This aggregation smooths transient fluctuations, although persistent assignment errors or limited prototype usage may still bias the estimate.

The estimated reliability is first used to calibrate prototype responses. Given the sparse response $\tilde{a}_{ik}$ produced by PTSL, PRSC obtains a reliability-aware response:

$$\tilde{a}^r_{ik} = \frac{\tilde{a}_{ik}\max(r_k, r_{\min})}{\sum_{j=1}^{M}\tilde{a}_{ij}\max(r_j, r_{\min}) + \epsilon}, \tag{18}$$

where M is the number of prototypes and $r_{\min}$ is a reliability floor. This operation suppresses responses assigned to unreliable prototypes while keeping the response distribution normalized. The reliability-aware response $\tilde{a}^r_{ik}$ is returned to PTSL and used as the effective response for prototype aggregation and residual adaptation. Therefore, PTSL refines target features according to the PRSC-calibrated prototype field rather than the raw sparse assignment alone.

PRSC further imposes reliability-aware weak–strong structure consistency. Let $\mathbf{q}^w_i = [q^w_{i1}, \ldots, q^w_{iM}]$ and $\mathbf{q}^s_i = [q^s_{i1}, \ldots, q^s_{iM}]$ denote the temperature-scaled assignment logits of the weak and strong views before top-k sparsification. The reliability-weighted structure consistency loss is defined as

$$\mathcal{L}_{rc} = \frac{1}{B}\sum_{i=1}^{B}\sum_{k=1}^{M} r_k \left(q^w_{ik} - q^s_{ik}\right)^2. \tag{19}$$

Different from directly minimizing the distance between weak and strong image features, this loss encourages the two views of the same target image to maintain consistent responses to the runtime prototypes. Reliable prototypes impose stronger consistency constraints, while unreliable prototypes contribute less.

The reliability score also modulates the prototype memory update in PTSL. Specifically, the update rate is defined as

$$\eta_k = \eta_{\min} + (\eta_{\max} - \eta_{\min}) r_k, \tag{20}$$

where $\eta_{\min} = 0.0005$ and $\eta_{\max} = 0.005$. For prototypes selected for update, η_k weights the current batch estimate and $1 - \eta_k$ the previous memory. Higher reliability favors new observations, whereas lower reliability retains more historical information.

Besides reliability-aware response consolidation, PRSC introduces active prototype de-redundancy to maintain diversity among frequently used prototypes. For prototype k, its response mass in the current target batch is computed from the strong-view sparse response:

$$S_k = \sum_{i=1}^{B}\tilde{a}^s_{ik}. \tag{21}$$

The active prototype set $\mathcal{A} \subseteq \{1, \ldots, M\}$ is selected as the top-M_a prototypes with the largest response mass:

$$\mathcal{A} = \mathrm{Top}_{M_a}\left(\{S_k\}_{k=1}^{M}\right), \tag{22}$$

where M_a is the number of selected active prototypes. The active prototype de-redundancy loss is applied only to the selected runtime prototypes:

$$\mathcal{L}_{apd} = \frac{1}{|\mathcal{A}|(|\mathcal{A}|-1)}\sum_{\substack{k,l\in\mathcal{A}\\ k\neq l}}\left[\max\left(0, \cos(p^{rt}_k, p^{rt}_l) - \delta\right)\right]^2, \tag{23}$$

where δ is the cosine similarity margin, fixed at 0.5 in our

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implementation. The loss penalizes pairwise cosine similarities above δ only among active runtime prototypes. Since these prototypes dominate the current target assignments, focusing the regularization on the active set reduces redundant structural responses in the current batch while leaving inactive prototypes unaffected. This encourages the prototype field to cover more diverse target structures.

Overall, PRSC complements PTSL from three aspects. First, it estimates prototype reliability from usage and weak–strong response consistency. Second, it feeds reliability-aware responses back to PTSL for prototype aggregation and residual adaptation, and uses reliability scores to modulate the prototype memory update rate. Third, it regularizes frequently activated runtime prototypes to reduce redundancy. Through these operations, PRSC consolidates the target prototype field and makes PTSL-based target feature refinement more stable during cross-domain representation learning.

E. Optimization and Target-domain Evaluation

The overall training objective consists of four parts: supervised source classification, target teacher–student consistency, reliability-aware weak–strong structure consistency, and active prototype de-redundancy. For labeled source samples, the student network is optimized by the standard cross-entropy loss:

$$\mathcal{L}_{ce} = \frac{1}{B_s} \sum_{i=1}^{B_s} \mathrm{CE}\left(h_\theta(f_\theta(x_i^s)), y_i^s\right), \tag{24}$$

where B_s is the source batch size, x_i^s is a source image, and y_i^s is its label.

For unlabeled target samples, the EMA teacher generates soft pseudo-labels from the weak target view, while the student predicts from the strong target view refined by PTSL with PRSC-calibrated prototype responses. Specifically, the teacher target is defined as

$$\mathbf{p}_i^t = \mathrm{Softmax}\left(\frac{h_{\theta'}(f_{\theta'}(x_i^{t,w}))}{T}\right), \tag{25}$$

where T is the teacher temperature. The teacher prediction is detached from gradient computation. Let $\tilde{z}_i^{t,s} = R_\psi(\tilde{u}_i^{t,s})$ denote the reconstructed representation of the strong target view. The target teacher–student consistency loss is computed as

$$\mathcal{L}_u = \frac{1}{B_t} \sum_{i=1}^{B_t} \mathrm{CE}\left(h_\theta(\tilde{z}_i^{t,s}), \mathrm{sg}(\mathbf{p}_i^t)\right), \tag{26}$$

where B_t is the target batch size and $\mathrm{sg}(\cdot)$ denotes the stop-gradient operation. In this loss, the target label is not used; the supervision signal is provided only by the EMA teacher.

The final objective is written as

$$\mathcal{L} = \lambda_{ce}\mathcal{L}_{ce} + \lambda_u\mathcal{L}_u + \lambda_{rc}\mathcal{L}_{rc} + \lambda_{apd}\mathcal{L}_{apd}, \tag{27}$$

where λ_{ce}, λ_u, λ_{rc}, and λ_{apd} balance source supervision, target teacher–student consistency, reliability-weighted structure consistency, and active prototype de-redundancy, respectively.

After each student update, the teacher parameters are updated by exponential moving average:

$$\theta' \leftarrow \mu\theta' + (1-\mu)\theta, \tag{28}$$

where μ is the EMA momentum. The EMA teacher is not optimized by back-propagation. It provides stable weak-view pseudo-labels, confidence gates, and teacher-side features for prototype memory update. This teacher–student design reduces the instability caused by directly using the current student prediction as the target supervision signal.

During training, each iteration proceeds as follows. First, the labeled source batch is used to compute $\mathcal{L}_{ce}$. Second, the weak and strong target views are processed: the weak view provides teacher pseudo-labels, confidence estimation, weak-side prototype responses, and memory-update features, while the strong view is refined by PTSL using PRSC-calibrated prototype responses and then used to compute $\mathcal{L}_u$. Third, PRSC computes $\mathcal{L}_{rc}$ from weak–strong prototype-field responses and applies $\mathcal{L}_{apd}$ to active runtime prototypes. Finally, all losses are combined to update the student, and the EMA teacher is updated from the student parameters.

After cross-domain representation learning, the learned model is evaluated under the standard support-query episodic few-shot setting. For each N-way K-shot episode, the support set D_T^S contains K labeled samples from each of the N target classes, and the query set D_T^Q contains samples from the same classes for evaluation. During this stage, the student backbone, PTSL, prototype memory, and PRSC components are kept fixed. Both support and query images are encoded by the frozen student feature extractor equipped with the learned target prototype field, where PTSL refines target features using the learned prototype memory and PRSC provides reliability-aware response weighting. A classifier is then adapted on the support features using their labels, and the adapted classifier is used to predict the query samples. The query labels are used only for computing classification accuracy. No backbone update, prototype memory update, reliability update, or structure-loss optimization is performed during few-shot evaluation.

IV. Experiments

In this section, we evaluate the proposed PARTS under cross-domain few-shot RS image classification settings. We first describe the benchmark datasets, evaluation protocols, and implementation details. Then, we compare PARTS with representative state-of-the-art methods using different source and target domains. After that, we conduct ablation experiments to verify the effectiveness of the main designs. We further investigate the sensitivity of key hyperparameters. Finally, visualization results are provided to intuitively demonstrate the effectiveness of the proposed method.

A. Datasets

PARTS is evaluated under cross-domain few-shot RS image classification settings. We use six representative datasets: NWPU-RESISC45 (NWPU) [47], EuroSAT [48], AID [49], UC Merced Land Use (UCM) [50], RSICB256 [51], and DIOR [52]. To evaluate robustness under different source-domain

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TABLE I: Detailed information about datasets used in experiments.

Dataset	Source	Resolution	Size	Category Number	Number Per Class
NWPU	Google Earth	0.2–30 m	256×256	45	700
EuroSAT	Sentinel-2 satellite	10 m	64×64	10	2000–3000
AID	Google Earth	0.5–8 m	600×600	30	220–420
UCM	USGS National Map Urban Area Imagery	1 ft (≈0.3 m)	256×256	21	100
RSICB256	Remote sensing images	0.3–3 m	256×256	35	≈690
DIOR	Optical remote sensing images	0.5–30 m	800×800	20	Imbalanced

TABLE II: Cross-domain 5-way classification performance under the 1-shot and 5-shot settings (source domain: NWPU).

Method	EuroSAT		DIOR		RSICB256		UCM		AID	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
MatchingNet	58.65 ± 0.91	79.49 ± 0.68	36.98 ± 0.78	57.28 ± 0.63	59.09 ± 1.06	79.36 ± 0.54	78.58 ± 0.87	89.16 ± 0.53	78.35 ± 0.90	87.73 ± 0.42
ProtoNet	60.85 ± 0.85	78.49 ± 0.65	53.28 ± 0.83	72.40 ± 0.72	77.43 ± 0.79	92.99 ± 0.41	76.32 ± 0.96	90.75 ± 0.64	76.94 ± 0.98	90.43 ± 0.59
RelationNet	55.12 ± 0.79	65.45 ± 0.61	49.72 ± 0.86	64.36 ± 0.58	73.85 ± 0.84	87.92 ± 0.47	75.68 ± 0.83	87.36 ± 0.59	76.56 ± 0.72	84.56 ± 0.46
SimCLR	60.53 ± 0.88	81.53 ± 0.60	41.36 ± 0.91	65.68 ± 0.64	65.42 ± 0.94	85.58 ± 0.55	63.21 ± 0.92	83.95 ± 0.68	67.53 ± 0.96	82.78 ± 0.72
FixMatch	68.53 ± 0.86	84.95 ± 0.58	56.74 ± 0.87	75.28 ± 0.52	82.36 ± 0.78	95.23 ± 0.43	83.98 ± 0.82	93.36 ± 0.42	81.51 ± 0.90	92.83 ± 0.46
LDP-Net	67.35 ± 1.02	82.06 ± 0.62	55.39 ± 0.90	75.90 ± 0.64	81.58 ± 0.76	94.57 ± 0.37	84.65 ± 0.90	93.30 ± 0.45	85.46 ± 0.87	93.47 ± 0.41
StyleAdv	73.52 ± 0.95	88.63 ± 0.55	64.01 ± 0.81	81.86 ± 0.51	88.03 ± 0.54	95.46 ± 0.35	85.06 ± 0.89	94.36 ± 0.36	79.51 ± 0.96	93.21 ± 0.40
DMN	68.34 ± 0.62	80.37 ± 0.62	54.18 ± 0.80	72.48 ± 0.53	80.26 ± 0.76	93.62 ± 0.40	74.90 ± 0.81	88.52 ± 0.68	80.30 ± 0.69	91.36 ± 0.36
UDL	73.41 ± 0.86	88.31 ± 0.56	61.42 ± 0.84	79.26 ± 0.36	86.87 ± 0.74	94.82 ± 0.37	84.25 ± 0.84	94.17 ± 0.38	82.35 ± 0.79	93.45 ± 0.41
MVP	60.12 ± 0.65	68.85 ± 0.57	48.76 ± 0.84	63.18 ± 0.56	74.92 ± 0.78	88.08 ± 0.48	78.78 ± 0.79	86.72 ± 0.50	70.53 ± 0.77	82.32 ± 0.57
BSCA	48.69 ± 0.85	71.55 ± 0.75	36.75 ± 0.89	56.64 ± 0.67	56.48 ± 0.91	78.38 ± 0.60	58.45 ± 0.92	78.71 ± 0.72	51.26 ± 0.86	72.26 ± 0.73
AFA	61.63 ± 0.94	82.49 ± 0.76	55.96 ± 0.87	75.63 ± 0.60	84.16 ± 0.78	93.52 ± 0.39	84.53 ± 0.78	93.46 ± 0.31	79.86 ± 0.81	93.63 ± 0.43
STARTUP	66.85 ± 0.84	84.05 ± 0.58	51.28 ± 0.83	72.47 ± 0.64	80.62 ± 0.87	92.75 ± 0.41	69.57 ± 0.91	90.02 ± 0.49	71.63 ± 0.89	89.54 ± 0.56
DDN	72.44 ± 0.88	88.82 ± 0.45	56.19 ± 0.90	77.51 ± 0.61	87.51 ± 0.70	95.23 ± 0.26	79.42 ± 0.85	93.13 ± 0.41	78.48 ± 0.84	92.76 ± 0.40
CDML	73.76 ± 0.73	88.32 ± 0.36	55.23 ± 0.83	74.86 ± 0.54	87.02 ± 0.71	95.13 ± 0.32	79.85 ± 0.87	91.72 ± 0.46	77.32 ± 0.90	89.92 ± 0.54
Ours (PARTS)	**79.14 ± 0.77**	**90.14 ± 0.39**	**65.84 ± 0.93**	**82.09 ± 0.60**	**89.12 ± 0.61**	**96.42 ± 0.29**	**85.23 ± 0.73**	**94.51 ± 0.41**	**85.54 ± 0.77**	**93.71 ± 0.42**

TABLE III: Cross-domain 5-way classification performance under the 1-shot and 5-shot settings (source domain: AID).

Method	EuroSAT		DIOR		RSICB256		UCM		NWPU	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
MatchingNet	58.37 ± 0.86	76.32 ± 0.65	42.98 ± 0.89	55.39 ± 0.58	62.00 ± 0.99	78.23 ± 0.45	66.75 ± 0.89	86.54 ± 0.55	62.16 ± 0.92	80.95 ± 0.62
ProtoNet	64.56 ± 0.93	76.83 ± 0.66	52.43 ± 0.86	69.54 ± 0.74	71.75 ± 0.84	91.67 ± 0.43	69.23 ± 0.91	85.36 ± 0.57	66.46 ± 0.86	81.94 ± 0.61
RelationNet	56.23 ± 0.81	64.38 ± 0.56	48.63 ± 0.79	63.96 ± 0.55	75.62 ± 0.86	87.82 ± 0.48	68.56 ± 0.87	84.27 ± 0.46	63.25 ± 0.66	79.26 ± 0.54
SimCLR	59.27 ± 0.89	78.93 ± 0.61	44.83 ± 0.88	65.12 ± 0.62	63.63 ± 0.89	85.76 ± 0.54	65.36 ± 0.86	79.63 ± 0.72	59.25 ± 0.94	75.90 ± 0.74
FixMatch	69.36 ± 0.83	82.31 ± 0.64	57.64 ± 0.83	75.46 ± 0.57	80.72 ± 0.74	92.64 ± 0.42	71.06 ± 0.87	89.05 ± 0.59	76.63 ± 0.79	86.16 ± 0.59
LDP-Net	66.32 ± 0.88	81.84 ± 0.60	57.92 ± 0.94	75.85 ± 0.65	82.34 ± 0.76	94.37 ± 0.35	75.55 ± 0.85	90.45 ± 0.50	70.61 ± 0.87	87.10 ± 0.55
StyleAdv	67.32 ± 0.82	85.23 ± 0.56	58.28 ± 0.84	76.73 ± 0.57	86.06 ± 0.69	94.88 ± 0.26	79.73 ± 0.81	92.03 ± 0.41	71.37 ± 0.85	89.26 ± 0.52
DMN	63.76 ± 0.63	78.05 ± 0.63	53.41 ± 0.76	70.64 ± 0.55	75.68 ± 0.74	90.86 ± 0.41	71.66 ± 0.85	81.57 ± 0.63	62.75 ± 0.86	73.65 ± 0.68
UDL	75.24 ± 0.82	88.23 ± 0.52	60.58 ± 0.84	75.22 ± 0.48	85.43 ± 0.71	94.42 ± 0.35	79.41 ± 0.84	91.86 ± 0.49	76.25 ± 0.87	90.28 ± 0.45
MVP	58.63 ± 0.54	71.96 ± 0.53	47.85 ± 0.75	62.34 ± 0.54	72.28 ± 0.73	86.84 ± 0.47	70.86 ± 0.49	82.09 ± 0.48	63.13 ± 0.65	76.46 ± 0.62
BSCA	54.28 ± 0.93	75.28 ± 0.75	39.64 ± 0.88	57.86 ± 0.66	61.85 ± 0.92	80.35 ± 0.59	56.94 ± 0.94	78.29 ± 0.71	55.76 ± 0.92	74.37 ± 0.75
AFA	58.37 ± 0.94	82.36 ± 0.75	52.76 ± 0.88	76.42 ± 0.59	76.94 ± 0.82	94.28 ± 0.40	71.53 ± 0.96	91.87 ± 0.44	65.28 ± 0.91	91.23 ± 0.58
STARTUP	64.15 ± 0.84	82.15 ± 0.67	53.68 ± 0.87	74.28 ± 0.67	78.38 ± 0.79	93.47 ± 0.35	70.42 ± 0.91	89.34 ± 0.52	70.68 ± 0.89	85.87 ± 0.56
DDN	72.35 ± 0.85	87.31 ± 0.51	57.10 ± 0.90	75.05 ± 0.63	85.86 ± 0.76	94.60 ± 0.31	76.18 ± 0.89	91.60 ± 0.45	74.55 ± 0.85	90.33 ± 0.45
CDML	73.76 ± 0.73	88.32 ± 0.36	55.23 ± 0.83	74.86 ± 0.54	87.02 ± 0.71	95.13 ± 0.32	**79.85 ± 0.87**	91.72 ± 0.46	69.56 ± 0.68	87.25 ± 0.53
Ours (PARTS)	**77.74 ± 0.78**	**89.30 ± 0.40**	**61.08 ± 0.89**	**77.81 ± 0.62**	**87.18 ± 0.69**	**95.22 ± 0.36**	79.52 ± 0.90	**92.16 ± 0.50**	**78.54 ± 0.87**	**91.43 ± 0.48**

conditions, NWPU and AID are separately used as labeled source domains, while the remaining five datasets in each setting are used as target domains. These two source domains are selected to represent different factors that may affect cross-domain transfer. NWPU provides a broader category space and more balanced source samples, allowing us to examine the effect of source-domain semantic coverage. In contrast, AID contains images collected from different sensors, regions, seasons, and imaging conditions, making it suitable for evaluating the influence of source-domain acquisition heterogeneity. The dataset statistics are summarized in Table I.

B. Experimental Setup

Following the unlabeled-target cross-domain few-shot evaluation protocol adopted in STARTUP [22] and DDN [23], we use all labeled source-domain samples and 20% of the target-domain images for cross-domain representation learning. The sampled target-domain images are used without labels and serve only as unlabeled target data. The remaining 80% of the target dataset is held out for episodic evaluation, and the unlabeled target subset is disjoint from all support and query images used during evaluation. During evaluation, 5-way 1-shot and 5-way 5-shot episodes are sampled from the held-

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out target subset; support labels are used only for episode-specific classifier adaptation, and query labels are used only for computing accuracy. The average top-1 accuracy over 600 randomly sampled episodes with 95% confidence intervals is reported as the evaluation metric.

For implementation, we adopt ResNet-10 [53] as the backbone, which is pre-trained on miniImageNet [9] to enhance feature representation. The model is trained for 40 epochs. For unlabeled target samples, each image is transformed into weak and strong augmented views for teacher–student consistency learning. The EMA teacher is updated from the student model with momentum $\mu = 0.99$ and is not optimized by back-propagation. In PTSL, the sparse prototype response keeps the top-6 prototypes. In the active prototype de-redundancy term, the top-8 activated prototypes are selected for regularization. The loss weights are set as $\lambda_{ce} = 1.0$, $\lambda_u = 1.0$, $\lambda_{rc} = 0.03$, and $\lambda_{apd} = 0.02$. During target-domain few-shot evaluation, the learned feature extractor is kept fixed, and only an episode-specific classifier is adapted on the support set before predicting the query samples. All experiments are implemented with PyTorch and conducted on a single NVIDIA RTX 3090 GPU.

C. Comparison With State-of-the-Art Methods

To comprehensively evaluate the proposed PARTS, we conduct experiments under two source-domain settings, where NWPU and AID are alternately used as the source domain, and the remaining RS datasets serve as target domains. The results under the 5-way 1-shot and 5-way 5-shot settings are reported in Tables II and III. For fair comparison, all methods use the same target-domain split and are evaluated on the same sampled episodes. Each compared method is implemented with its recommended or best-performing configuration, and methods requiring unlabeled target data use the same unlabeled subset D_T^U.

The compared methods cover three categories, including meta-learning-based few-shot methods, transfer-learning-based CDFSL methods, and related RS methods. Specifically, we compare PARTS with representative few-shot methods such as MatchingNet [9], ProtoNet [10], and RelationNet [44], as well as recent CDFSL methods including AFA [54], StyleAdv [13], LDP-Net [12], STARTUP [22], and DDN [23]. Generic self-supervised and semi-supervised baselines, including SimCLR [55] and FixMatch [39], are also included. In addition, several remote-sensing-oriented methods, including BSCA [40], MVP [16], DMN [15], UDL [25], and CDML [32], are included for comparison. Since CDML follows its original multi-source training protocol, its results on the same target domains remain unchanged across different source-domain settings.

With NWPU as the source domain, PARTS achieves average accuracies of 80.97% and 91.37% under 1-shot and 5-shot evaluation, respectively. The gains are more pronounced in some low-shot settings: on EuroSAT, the 1-shot accuracy exceeds the strongest compared result by 5.38 percentage points. However, the improvements are not uniform. In the 5-shot setting, the gains over the strongest compared methods are only 0.23 and 0.15 percentage points on DIOR and UCM, respectively. These results indicate that the benefit of prototype-guided refinement varies with the target dataset and the amount of labeled support available during evaluation.

To further verify the generalization ability of PARTS, we also conduct experiments using AID as the source domain. PARTS achieves the best average performance under both evaluation settings, with average accuracies of 76.81% and 89.18% in the 1-shot and 5-shot settings, respectively. In the 1-shot setting, PARTS ranks first on most target domains, while CDML obtains the best result on UCM. In the 5-shot setting, PARTS achieves the best result across all five target domains, further demonstrating its effectiveness under different source-domain conditions.

Compared with conventional metric-based few-shot methods, PARTS achieves substantial improvements, suggesting that directly transferring a source-trained metric space is insufficient for remote sensing CDFSL under large domain shifts. PARTS also outperforms generic self-supervised and semi-supervised baselines such as SimCLR and FixMatch, suggesting that contrastive representation learning or prediction-level consistency alone is insufficient to explicitly model target-domain prototype structures. Moreover, the competitive average performance against recent remote-sensing-oriented methods further supports the effectiveness of explicitly modeling adaptive target prototype structures.

The competitive performance of PARTS can be attributed to its target-structure-aware design. Instead of relying only on source-domain decision boundaries or prediction-level alignment, PARTS constructs an adaptive prototype field for target-domain samples and refines target features according to their prototype responses. The reliability-aware structure consolidation mechanism further suppresses unreliable prototype responses and encourages stable weak–strong target structures. These designs enable PARTS to learn more transferable and target-discriminative representations under different cross-domain few-shot settings.

D. Ablation Experiments

To examine the effect of each design in the proposed framework, ablation experiments are conducted with NWPU as the source domain and five datasets as target domains, including EuroSAT, DIOR, RSICB256, UCM, and AID. Table IV reports the results under the 5-way 1-shot setting. The ablation starts from a source-supervised baseline and gradually incorporates teacher–student learning, PTSL, and PRSC. Here, teacher–student learning denotes the basic target-domain consistency branch, where the EMA teacher generates pseudo labels from weakly augmented target images to supervise the student predictions on strongly augmented target images, without introducing the target prototype field.

The source-supervised baseline achieves an average accuracy of 68.02%, indicating that direct transfer from the source domain is still limited under large domain shifts. After introducing teacher–student learning, the average accuracy increases to 73.09%. This improvement shows that unlabeled target data provide useful adaptation cues, but the model still relies mainly on sample-level consistency and lacks an explicit mechanism to organize target-domain structures. When

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TABLE IV: Ablation study of the proposed method under the 5-way 1-shot setting.

Method	T-S	PTSL		PRSC		5-way 1-shot Acc. (%)					
		Prototype	Memory	Rel.	De-red.	EuroSAT	DIOR	RSICB256	UCM	AID	Average
1 Baseline	×	×	×	×	×	63.63	52.94	76.42	74.13	72.98	68.02
2	✓	×	×	×	×	71.22	53.48	85.68	78.16	76.89	73.09
3	✓	✓	×	×	×	74.08	58.36	86.72	80.52	79.64	75.86
4	✓	✓	✓	×	×	75.66	60.74	87.54	82.08	81.86	77.58
5	✓	✓	✓	✓	×	77.38	63.18	88.42	83.52	83.74	79.25
6	✓	✓	✓	×	✓	77.02	62.54	88.66	83.10	83.28	78.92
7 Ours	✓	✓	✓	✓	✓	**79.14**	**65.84**	**89.12**	**85.23**	**85.54**	**80.97**

TABLE V: Computational overhead of the ablation configurations under the NWPU-to-EuroSAT setting.

Method	Params. (M)	Time (s/epoch)	Peak GPU memory (MiB)	Inference (ms/image)
1 Baseline	4.929	9.60	3404	1.364
2 T-S	4.952	20.50	4566	1.453
3 T-S + Prototype	6.093	31.63	4578	4.213
4 T-S + Prototype + Memory	6.093	36.20	4578	4.392
5 T-S + Prototype + Memory + Rel.	6.093	49.08	4580	4.261
6 T-S + Prototype + Memory + De-red.	6.093	43.85	4578	4.437
7 Ours	6.093	51.65	4580	4.344

TABLE VI: Backbone comparison with NWPU as the source domain.

Backbone	Method	Avg. Acc. (%)	
		1-shot	5-shot
ResNet-10	T-S	73.09	82.31
	PARTS	80.97	91.37
ResNet-18	T-S	75.98	84.28
	PARTS	85.03	92.16

the target prototype field is further introduced, the average performance rises to 75.86%, suggesting that prototype-based target structure modeling is beneficial for cross-domain representation learning. The prototype memory brings another gain, improving the average accuracy to 77.58%. Compared with using only the instantaneous prototype field, memory update provides a more stable target-domain structural reference during training. On this basis, the two components of PRSC are evaluated separately. Reliability modeling improves the average result to 79.25%, while active prototype de-redundancy reaches 78.92%. These results indicate that both reliability-aware response consolidation and redundancy suppression among active prototypes contribute to stabilizing the target prototype field, but they emphasize different aspects of structure refinement.

With all components enabled, the complete model obtains the best average accuracy of 80.97%, outperforming the source-supervised baseline by 12.95 percentage points and the teacher–student baseline by 7.88 percentage points. The consistent improvement across different target domains demonstrates that PTSL and PRSC are complementary: PTSL builds an adaptive prototype field for target samples, whereas PRSC further consolidates this field by selecting reliable prototype responses and reducing redundant active prototypes.

Table V summarizes the computational costs of the ablation configurations. Prototype adaptation introduces additional parameters and inference computation, whereas memory, reliability estimation, and de-redundancy add no further parameters at the reported precision. PARTS requires more training and inference time than the teacher–student baseline, while peak GPU memory remains nearly unchanged. To further examine its applicability to a larger backbone, Table VI reports results with ResNet-18 using NWPU as the source domain. PARTS improves average accuracy over the corresponding baseline by 9.05 and 7.88 percentage points under 1-shot and 5-shot evaluation, respectively, showing that its benefits extend beyond ResNet-10.

E. Parameter Analysis

We analyze two key hyperparameters in PTSL: the number of target prototypes M and the top-k sparse response size. Fig. 5 reports average 5-way 1-shot and 5-shot accuracy over the corresponding target datasets, using NWPU and AID as source domains. The prototype field captures the distribution of unlabeled target features, with M controlling the granularity of the structural representation. Under both source-domain settings, average accuracy increases as M grows from 16 to 32 and decreases thereafter in both shot protocols. A small prototype set may provide insufficient coverage of target structures, whereas excessive prototypes may introduce redundancy and distribute assignment mass across too many

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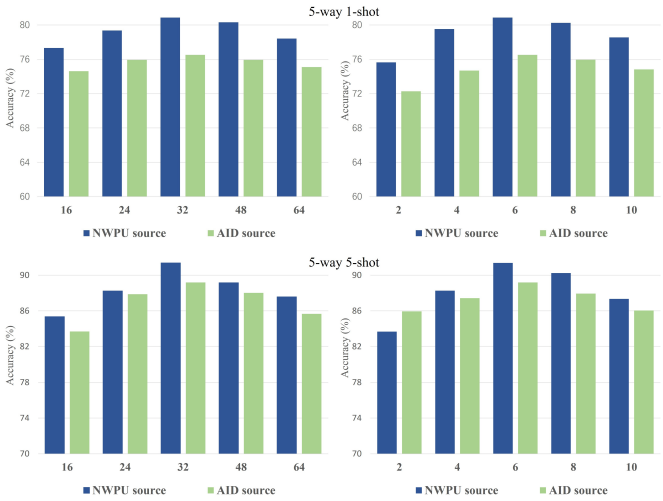


Fig. 5. Parameter sensitivity analysis of PTSL. The left and right columns show the effects of the number of target prototypes M and the top-k sparse response size, respectively. The top and bottom rows report average 5-way 1-shot and 5-shot accuracy over the corresponding target datasets, using NWPU and AID as source domains.

TABLE VII: Loss-weight sensitivity averaged over target datasets with NWPU as the source.

λ_{rc}	λ_{apd}	1-shot Acc. (%)	5-shot Acc. (%)
Varying λ_{rc} with $\lambda_{apd} = 0.02$			
0.01	0.02	78.98	89.23
0.02	0.02	79.36	90.47
0.03	**0.02**	80.97	91.37
0.05	0.02	79.61	89.96
Varying λ_{apd} with $\lambda_{rc} = 0.03$			
0.03	0.005	79.37	89.68
0.03	0.01	80.25	90.89
0.03	**0.02**	80.97	91.37
0.03	0.04	79.86	90.43

anchors. The consistent average peak at $M = 32$ motivates its use as a shared default across the evaluated source–target configurations.

A similar trend is observed for the top-k sparse response size, with $k = 6$ achieving the highest average accuracy under both 1-shot and 5-shot evaluation for both source domains. When k is too small, the assignment becomes overly sharp and may discard useful local structural information. Increasing k to 6 improves the performance by maintaining a balance between prototype competition and structural coverage. However, larger values of k involve more prototypes in the response, which may weaken the selectivity of prototype assignment and slightly degrade performance. Therefore, we set $M = 32$ and $k = 6$ as the default configuration.

We further examine the loss weights λ_{rc} and λ_{apd} by varying one while fixing the other at its default value. Table VII reports average accuracy across target datasets with NWPU as the source domain. The combination $\lambda_{rc} = 0.03$ and $\lambda_{apd} = 0.02$ achieves the highest average accuracy in both shot settings. Across the tested values, accuracy varies by up to 2.14 percentage points for λ_{rc} and 1.69 percentage points for λ_{apd}. Both smaller and larger weights yield lower accuracy than the default combination, indicating the importance of balancing the auxiliary objectives with the classification losses.

F. Visualization Analysis

1) Feature Embedding Visualization (t-SNE): To further analyze the representation quality learned by the proposed method, we visualize target-domain feature embeddings using t-SNE [56], as shown in Fig. 6. Compared with the baseline, the proposed method produces more compact and better separated feature distributions on different target domains. In the baseline results, samples from different categories are often mixed together, and several clusters are scattered or connected by ambiguous transition regions. This indicates that sample-level consistency alone is insufficient to organize target-domain samples into stable class-aware structures under domain shift.

After introducing PTSL and PRSC, the target features show clearer local grouping and more distinguishable class boundaries. PTSL constructs an adaptive target prototype field to guide target features toward prototype-aligned representations, while PRSC further consolidates reliable prototype responses and suppresses redundant active prototypes. As a result, intra-class samples become more compact, and inter-class overlap is reduced. Although some visually similar categories still exhibit partial mixing in challenging target domains, the overall feature structure is more organized than that of the baseline. These observations are consistent with the quantitative results and demonstrate that the proposed method improves target-domain feature separability by modeling reliable prototype-level structures.

2) Grad-CAM Visualization: Fig. 7 compares Grad-CAM [57] visualizations of the baseline and PARTS using identical held-out target images, with each column corresponding to the same input across methods. The examples cover several scene categories, including rivers, roads, runways, and aircraft, illustrating model responses to different spatial structures. The first row shows the original images, while the second and third rows show the corresponding heatmaps of the baseline and the proposed method, respectively. Compared with the baseline, our method produces more focused and semantically meaningful activation regions. The baseline often responds to broad background areas or irrelevant surrounding textures, indicating that its target-domain representation is still affected by domain shift. In contrast, the proposed method tends to highlight more discriminative regions, such as river bodies, roads, runways, and aircraft structures. These responses are more consistent with the visual semantics of the target categories. This comparison suggests that PTSL and PRSC not only improve the separability of target-domain features, but also help the model focus on more reliable visual evidence, providing qualitative support for the effectiveness of prototype-level target structure modeling.

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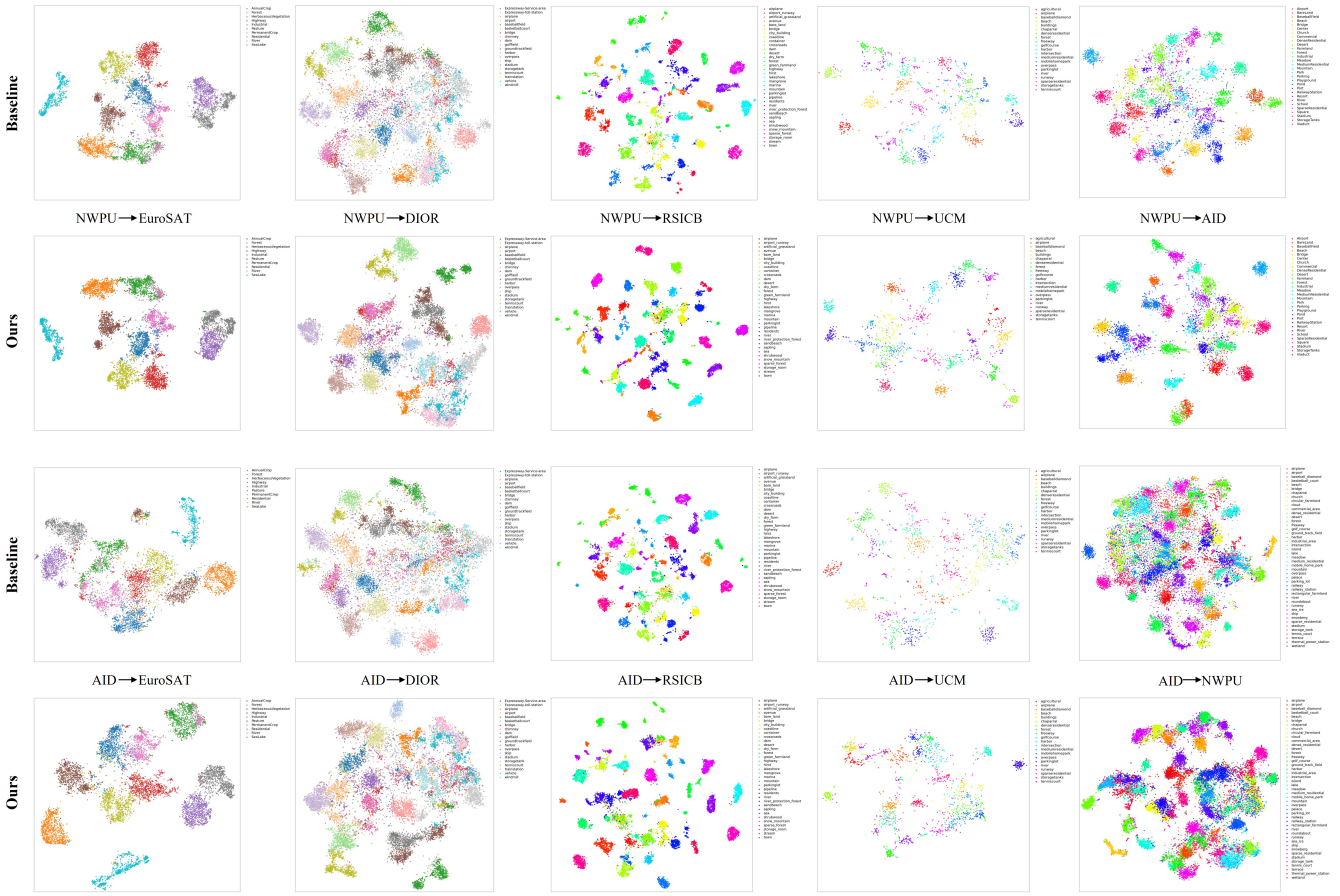


Fig. 6. t-SNE visualization of target-domain feature embeddings. The proposed PARTS method yields more compact clusters and clearer class boundaries than the baseline.

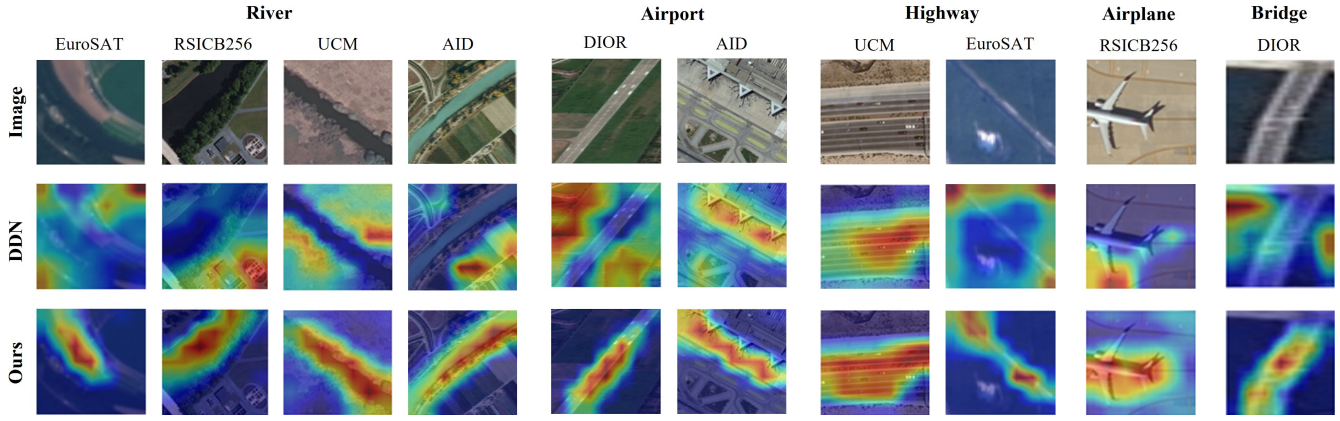


Fig. 7. Grad-CAM visualization on the same selected held-out target-domain images. Compared with the baseline, the proposed PTSL+PRSC method attends more accurately to discriminative semantic regions.

V. Conclusion

In this article, we proposed PARTS for cross-domain few-shot remote-sensing image classification. The method consists of two main components: PTSL and PRSC. PTSL constructs an adaptive target prototype field from unlabeled target-domain samples, where target features are assigned to runtime prototypes and refined through prototype-guided residual adaptation. PRSC further consolidates this field by estimating prototype reliability and reducing redundancy among active prototypes, so that the learned target representation becomes more stable and discriminative under domain shift. The proposed design complements teacher–student consistency by introducing prototype-guided target feature refinement.

Extensive experiments on multiple cross-domain remote-sensing benchmarks demonstrate the effectiveness of the proposed method under both 1-shot and 5-shot settings. Ablation studies and visualization results further confirm the effectiveness and interpretability of the proposed design. Overall, the proposed framework effectively exploits unlabeled target data to bridge domain gaps and enhance few-shot transferability. Future work will explore more efficient prototype updating strategies and extend the proposed prototype-level structure

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modeling to other remote-sensing tasks such as object detection and semantic segmentation.

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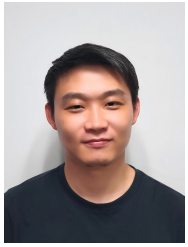
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